ModelIt!

Student Activity Pack

Hand Warmers and Hidden Reactions

How Atoms Rearrange to Create Heat, Light, and Everything You Use

Grade 8 | NGSS MS-PS1-4, MS-PS1-5 | Energy and Matter

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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Table of Contents

ModelIt!

1 Pre-Assessment 3

2 Scoring Rubric 4

3 Component Analysis 5

4 Model Recording Page 6

5 Simulation Observations 7

6 Research & Extend 8

7 Reflection Questions 9

8 STEM Challenge 10

1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 How do you think a hand warmer works?

2 What is a chemical reaction?

3 What happens to the atoms during a chemical reaction?

4 If you weigh the ingredients before a reaction and the products after, will the mass be different?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Iron Powder Amount | Oxygen Exposure | Reaction Rate | Heat Output

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Sealed (No Oxygen)

Iron powder present, no oxygen

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Normal Use

Normal iron amount, moderate oxygen

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Maximum Exposure

Maximum iron, maximum oxygen

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Salt (Catalyst) — Salt speeds up the iron-oxygen reaction without being consumed

Carbon (Adsorbent) — Activated carbon helps distribute heat evenly and holds moisture

Insulation — The packaging material that controls how quickly heat escapes

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

How can a small packet create heat without fire, batteries, or electricity? Use evidence from your simulation to support your answers.

1 Why does a hand warmer stop working eventually? Where did the iron go?

2 How is rusting a nail the same chemical reaction as a hand warmer, just slower?

3 If atoms aren't created or destroyed, why does the hand warmer feel like it 'created' heat?

4 What's the difference between a hand warmer (exothermic) and an instant cold pack (endothermic)?

5 How does controlling oxygen exposure control the reaction rate?

8 STEM Challenge: Design a Better Hand Warmer

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YOUR ENGINEERING MISSION

Design an improved hand warmer that provides consistent heat for a specific duration using your understanding of reaction rates and energy transfer.

THE SCENARIO

An outdoor equipment company wants to create a hand warmer that lasts exactly 8 hours at a comfortable temperature for winter hikers. Use your model to design the specifications.

Engineering Design Thinking

1. HOW MUCH IRON POWDER WILL YOU USE?
2. HOW WILL YOU CONTROL OXYGEN FLOW TO REGULATE TEMPERATURE?
3. WHAT INSULATING MATERIALS WILL YOU USE?
4. HOW WILL YOU VERIFY THAT MASS IS CONSERVED IN YOUR DESIGN?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Chemical Engineers design products using chemical reaction principles. Materials Scientists develop new materials with specific thermal properties. They earn $75,000-$130,000/year.